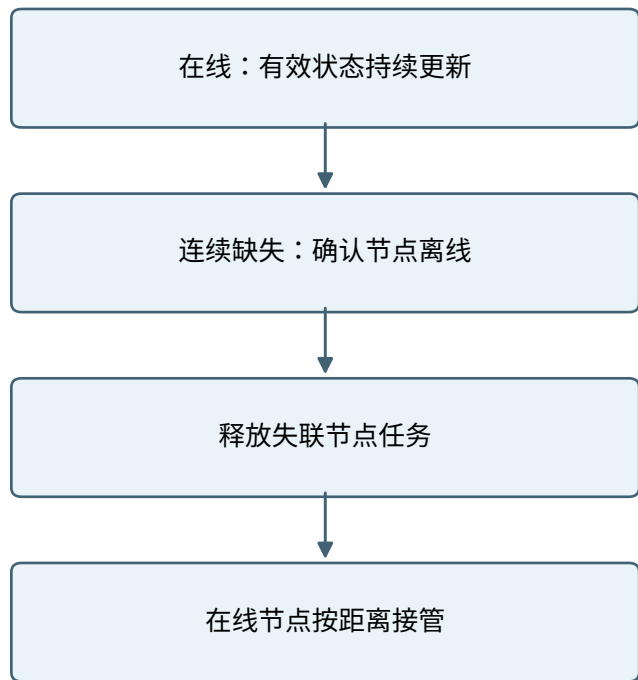
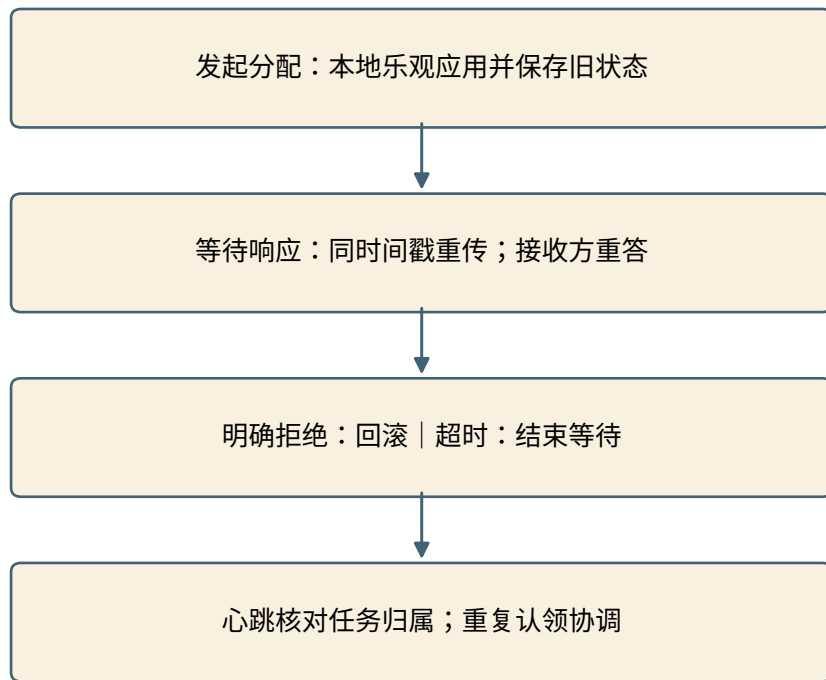


(a) 节点掉线 → 任务接管 (机制图)



(b) 任务层丢包 → 协调 (机制图)



20% 是任务消息注入条件；不丢弃 swarm_traj。状态箭头不表示实测时延或成功概率。